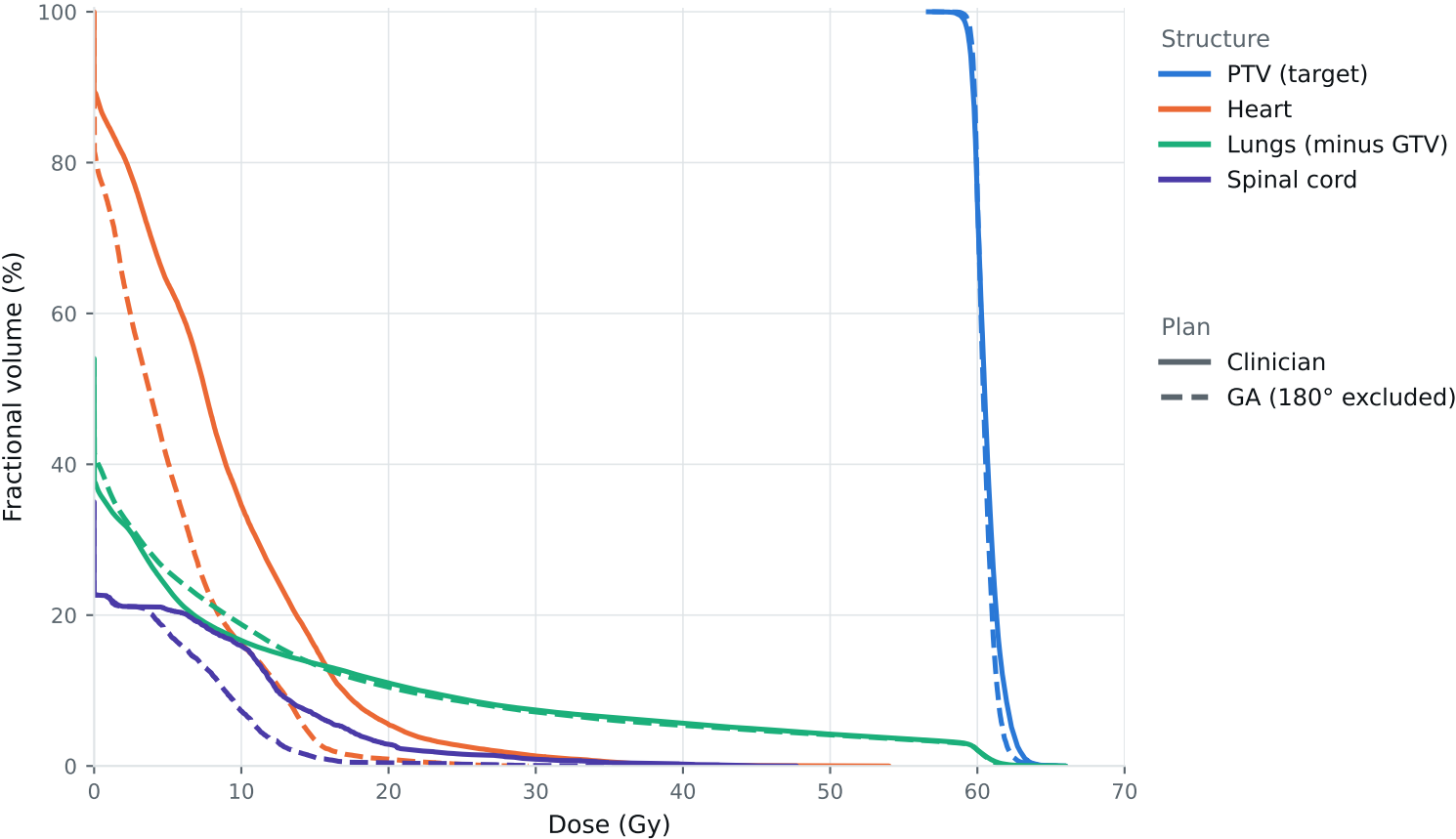


# Dose-volume histogram — Lung\_Patient\_28

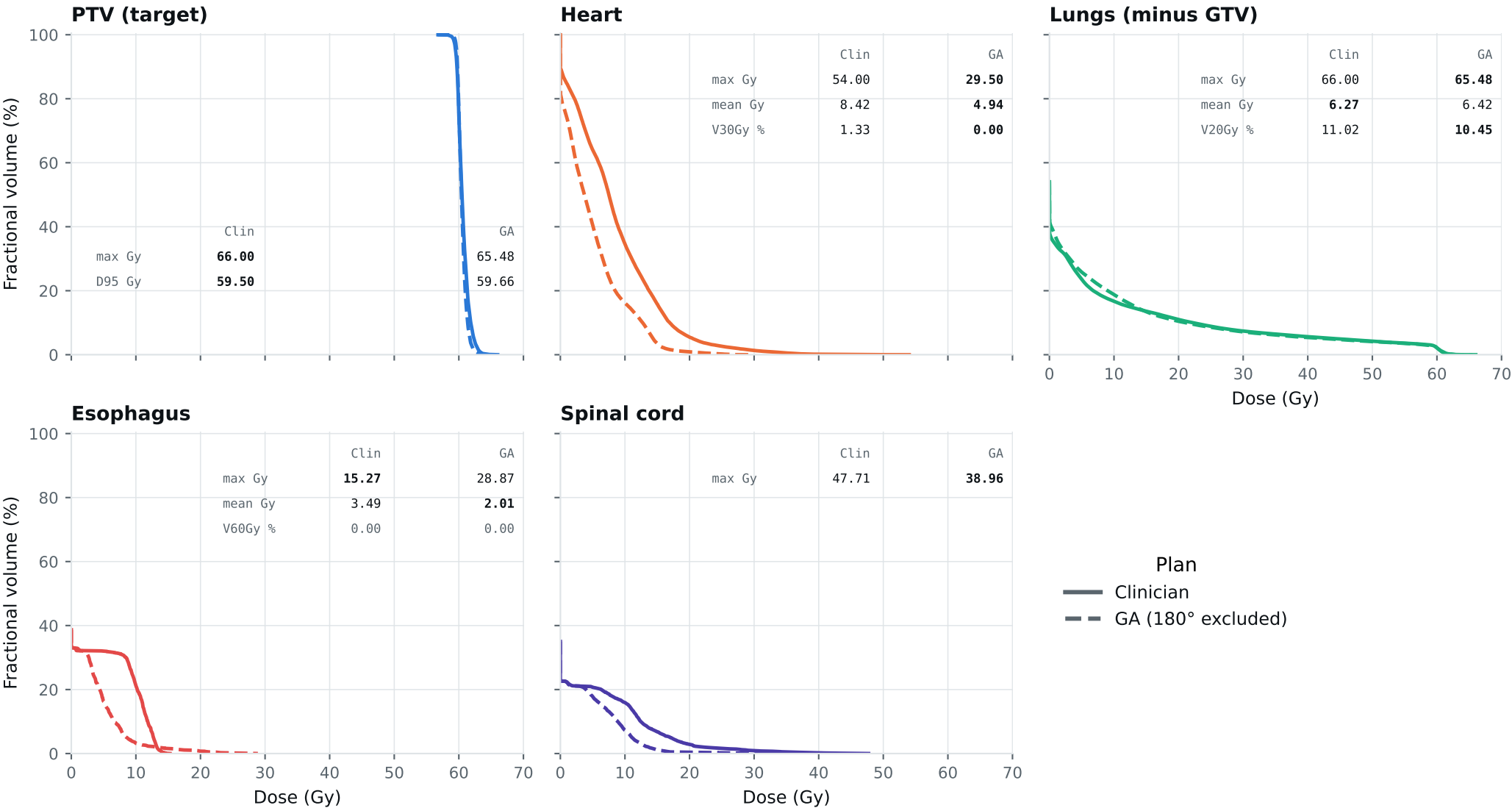
Clinician plan vs. GA (180° excluded). A curve further left and lower means less dose to that structure.



Curves further left are better for organs; the PTV curve should stay high, then drop sharply. Target curves should overlap; organ curves show the tradeoffs. Esophagus is on page 2.

# DVH by structure — Lung\_Patient\_28

Same curves, one structure per panel, all plans, each annotated with that structure's protocol metrics.  
Bold = clinically preferable: organs favor lower dose; PTV favors the value closest to goal within the limit. Grey = all plans are within 0.01 of each other. Amber misses a goal,



# Clinical criteria — Lung\_Patient\_28

Organs: lower is better. PTV: closest to goal without exceeding the limit. Bold marks the best plan.

| Criterion                       | Limit     | Goal                                    | Clinician    | GA<br>180° excluded |    |
|---------------------------------|-----------|-----------------------------------------|--------------|---------------------|----|
| GTV max                         | 69        | 66                                      | 64.32        | <b>62.96</b>        | Gy |
| PTV max                         | 69        | 66                                      | <b>66.00</b> | 65.48               | Gy |
| ESOPHAGUS max                   | 66        | —                                       | <b>15.27</b> | 28.87               | Gy |
| ESOPHAGUS mean                  | 34        | 21                                      | 3.49         | <b>2.01</b>         | Gy |
| ESOPHAGUS V60Gy                 | 17        | —                                       | 0.00         | 0.00                | %  |
| HEART max                       | 66        | —                                       | 54.00        | <b>29.50</b>        | Gy |
| HEART mean                      | 27        | 20                                      | 8.42         | <b>4.94</b>         | Gy |
| HEART V30Gy                     | 50        | 48                                      | 1.33         | <b>0.00</b>         | %  |
| LUNG_L max                      | 66        | —                                       | <b>30.38</b> | 42.37               | Gy |
| LUNG_R max                      | 66        | —                                       | 66.00        | <b>65.48</b>        | Gy |
| CORD max                        | 50        | 48                                      | 47.71        | <b>38.96</b>        | Gy |
| SKIN max                        | 60        | —                                       | <b>51.86</b> | 52.33               | Gy |
| LUNGS_NOT_GTV max               | 66        | —                                       | 66.00        | <b>65.48</b>        | Gy |
| LUNGS_NOT_GTV mean              | 21        | 20                                      | <b>6.27</b>  | 6.42                | Gy |
| LUNGS_NOT_GTV V20Gy             | 37        | —                                       | 11.02        | <b>10.45</b>        | %  |
| PTV D95 (target coverage)       |           |                                         | 59.50        | 59.66               |    |
| Objective value (search signal) |           |                                         | 72.33        | 58.32               |    |
| MOSEK solve time                |           |                                         | 19 s         | 24 s                |    |
| Beam angles (gantry°)           | Clinician | 160°, 0°, 330°, 305°, 245°, 215°, 185°  |              |                     |    |
|                                 | GA        | 305°, 245°, 185°, 30°, 165°, 270°, 285° |              |                     |    |

Grey rows: all plans agree within 0.01 — heart, lung and LUNGS\_NOT\_GTV max sit at 66 Gy because those structures abut the target. Amber misses the goal; red exceeds the limit.

Source: Lung\_Patient\_28\_clinical\_metrics\_full\_resolution.json, full resolution. The curves and table come from the same solve.